Package ‘rENM’

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Type Package

Title Top-level orchestration for the rENM Framework

Version 0.1.0.9000

Description Provides the top-level rENM() function, which executes the complete retrospective ecological niche modeling pipeline for a single species by orchestrating the installed rENM Framework packages in sequence -- from occurrence extraction and environmental variable preparation through ensemble modeling, suitability trend analysis, AI-assisted interpretation, and final report assembly.

Depends R (>= 4.1.0)

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Encoding UTF-8

Roxygen list(markdown = TRUE)

URL https://github.com/rENM-Framework/rENM

BugReports https://github.com/rENM-Framework/rENM/issues

Imports rENM.core,
rENM.data,
rENM.model,
rENM.analysis,
rENM.ai,
rENM.reports,
utils

Config/roxygen2/version 8.0.0

Contents

`rENM` *Run the end-to-end rENM pipeline for a single bird species*

Description

Executes the full retrospective ecological niche modeling (rENM) workflow for a single species identified by its four-letter bird banding code. The function wraps the major data acquisition, occurrence processing, variable preparation, time-series modeling, trend analysis, AI reporting, and final report assembly steps implemented across the rENM framework packages.

Usage

```
rENM(alpha_code)
```

Arguments

`alpha_code`	Character scalar. The four-letter bird banding code for the target species. The value is normalized to uppercase before processing.

Details

This function serves as the top-level orchestration wrapper for the installed rENM framework packages:

- `rENM.core`
- `rENM.data`
- `rENM.model`
- `rENM.analysis`
- `rENM.ai`
- `rENM.reports`

For the supplied species alpha code, the function runs the complete pipeline in sequence, including:

1. extraction and preparation of eBird occurrence records,
2. spatial thinning and record limiting,
3. range-based extent determination,
4. extraction and staging of MERRA environmental variables,
5. stochastic variable screening,
6. construction of a 5-year-binned rENM time series,
7. climatic suitability trend analyses,
8. centroid, velocity, and hotspot analyses,
9. compilation of report tables, pages, and summary graphics,
10. assembly and submission of an AI-ready suitability package,
11. rendering of returned AI documents, and
12. assembly of the final report.

The function time-stamps the start and end of the run, computes total elapsed time, displays these values on the console, and appends them to the species log file at:

```
<project-dir>/runs/<alpha_code>/_log.txt
```

The project directory is resolved automatically by each framework package via `rENM.core::rENM_project_dir()`, which reads from the `rENM.project_dir` option or the `RENM_PROJECT_DIR` environment variable.

Log entries are written in the framework's standard timestamped format and bracketed with 72-character separator lines for readability.

The function uses `on.exit()` to guarantee that end time and elapsed time are recorded even if the pipeline terminates early, and `tryCatch()` to log any error before re-throwing it.

Value

Invisibly returns a named list containing:

- `alpha_code`: normalized four-letter species code,
- `start_time`: POSIXct start time,
- `end_time`: POSIXct end time,
- `elapsed_time`: difftime object giving total elapsed time,
- `log_file`: full path to the run log file,
- `status`: character string giving the final run status.

The primary purpose of the function is side-effect execution of the complete rENM workflow and generation of its associated outputs.

See Also

`rENM.core::rENM_project_dir`

Examples

```
## Not run:
rENM("CASP")

## End(Not run)
```

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